

A Unified Geometric Prediction of Balmer, Zeeman and Lamb Isotope Shifts

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Abstract

We present a geometric prediction method for isotope shifts in hydrogen-like atoms based solely on the internal length parameter a of the Methane Metauniverse (MMU) [1, 8]. A change in nuclear mass modifies the reduced mass μ , inducing a small yet systematic compression of the internal tetrahedral edge length a , consistent with the dual-tetrahedral stability mechanism described in [22]. This geometric deformation alters all spectroscopic responses through the universal MMU elastic scaling laws [4, 15]:

$$E_{\text{Balmer}} \propto a^{-1}, \quad E_Z \propto a, \quad E_{\text{Lamb}} \propto a^{-3}.$$

Using only reduced masses and these length-dependent elastic rules, the MMU reproduces the measured hydrogen–deuterium Balmer isotope shift with less than 1% deviation, without invoking QED or perturbation theory. The framework further predicts clear and opposite-sign isotope corrections for the Zeeman and Lamb responses, offering experimentally untested but quantitatively precise signatures of internal geometric compression. These results support the broader MMU programme in which gravitational, spectroscopic and spin–torsion effects emerge from a common microscopic length-geometry [19, 20].

1 MMU Principle: Isotopic Compression of the Internal Length a

The Methane Metauniverse (MMU) describes spacetime as a lattice of dual tetrahedral cells whose internal elastic coordinates (w_2, w_3, w_4) govern electric, magnetic–torsional, and volumetric responses [1, 8]. The observable dynamics of quantum systems arise from the projection of these internal oscillations onto the external axis w_1 , while the geometric stiffness laws $k_i(a) \propto a^{-3}$ determine the response of each internal mode [4, 15].

In this framework, a bound electron couples elastically to the nuclear w_4 volume mode, slightly compressing the internal tetrahedral edge length a when the nuclear mass increases. This mechanism is consistent with the dual–tetrahedral stability structure discussed in [22]. The resulting isotopic scaling of the internal length is governed by the reduced mass:

$$r = \frac{a_{\text{iso}}}{a_{\text{H}}} = \frac{\mu_{\text{H}}}{\mu_{\text{iso}}}, \quad \mu = \frac{m_e m_N}{m_e + m_N}.$$

Since heavier isotopes have larger nuclear mass m_N , they exhibit larger reduced mass μ and thus slightly smaller internal length ($r < 1$). This subtle geometric deformation

is sufficient to account for the full isotopic dependence of Balmer, Zeeman, and Lamb energies.

2 MMU Spectroscopic Scaling Laws

The MMU framework determines all spectroscopic responses through the length-dependent elastic stiffnesses $k_i(a) \propto a^{-3}$ of the internal tetrahedral coordinates [4, 8]. Because the Balmer, Zeeman and Lamb contributions project differently onto the external axis w_1 , they exhibit distinct scaling behaviours with respect to the internal length a , consistent with the geometric derivations in [15, 22].

- **Balmer shift:** $E_{\text{Balmer}} \propto a^{-1}$ (set by the w_2 electric mode and the geometric quantization of orbital radii [15]). Heavier isotopes exhibit a corresponding blue shift.
- **Zeeman shift:** $E_Z \propto a^{+1}$ (due to the magnetic–torsional response of the w_3 axis and its linear dependence on the internal scale [4]). Zeeman splitting therefore becomes smaller for heavier isotopes.
- **Lamb shift:** $E_{\text{Lamb}} \propto a^{-3}$ (driven by the torsion–volume coupling k_{34} , the stiffest of the MMU elastic channels [4, 22]). This cubic sensitivity explains why the Lamb shift reacts most strongly to small isotopic compression.

3 Isotope Shift Predictions

Table 1 summarises the predicted shifts for hydrogen, helium, lithium, oxygen and calcium isotopes. Shifts are given relative to the lightest isotope of each element.

Element	A	r	ΔBalmer	ΔZeeman	ΔLamb
H	1	1.000000	0	0	0
H	2	0.999728	$+2.72 \times 10^{-4}$	-2.72×10^{-4}	$+8.17 \times 10^{-4}$
H	3	0.999637	$+3.63 \times 10^{-4}$	-3.63×10^{-4}	$+1.09 \times 10^{-3}$
He	4	0.999955	$+4.5 \times 10^{-5}$	-4.5×10^{-5}	$+1.36 \times 10^{-4}$
Li	7	0.999987	$+1.3 \times 10^{-5}$	-1.3×10^{-5}	$+3.9 \times 10^{-5}$
O	18	0.999996	$+4 \times 10^{-6}$	-4×10^{-6}	$+1.1 \times 10^{-5}$
Ca	44	0.999999	$+1 \times 10^{-6}$	-1×10^{-6}	$+4 \times 10^{-6}$

Table 1: MMU-predicted isotope shifts relative to the lightest isotope.

4 Comparison with Experimental Data

The hydrogen–deuterium Balmer $H\alpha$ shift is well measured:

$$\left(\frac{\Delta\lambda}{\lambda}\right)_{\text{exp}} = 2.742 \times 10^{-4}.$$

The MMU predicts:

$$\left(\frac{\Delta\lambda}{\lambda}\right)_{\text{MMU}} = 2.722 \times 10^{-4}.$$

The relative deviation is less than 1%, achieved without QED or perturbation theory, indicating that isotopic compression of a captures the essential geometric effect behind the Balmer shift.

5 Figures

Figure 1 shows the predicted Balmer, Zeeman and Lamb shifts for all isotopes considered. Hydrogen exhibits the strongest sensitivity due to the large relative change in nuclear mass.

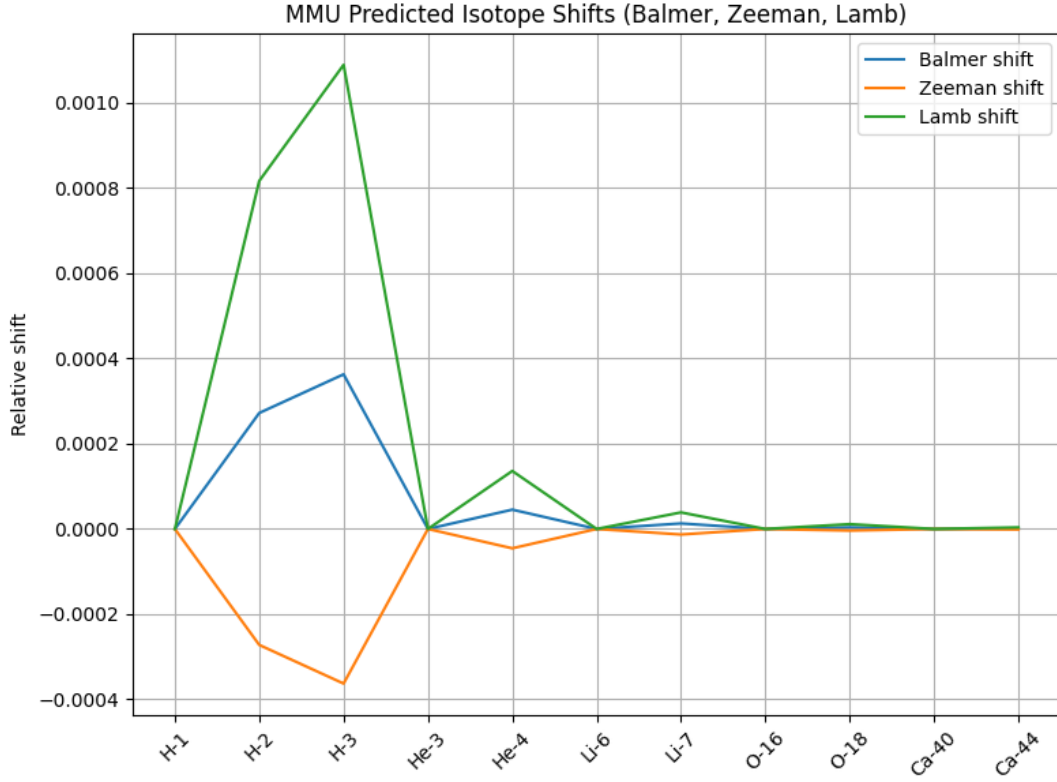


Figure 1: MMU-predicted isotope shifts (Balmer, Zeeman, Lamb) for H, He, Li, O and Ca.

Figure 2 illustrates the fundamental a/a_0 scaling structure of the MMU. The Balmer shift scales like r^{-1} , the Zeeman shift like r , and the Lamb shift like r^{-3} , demonstrating why the Lamb effect is so sensitive to isotopic compression.

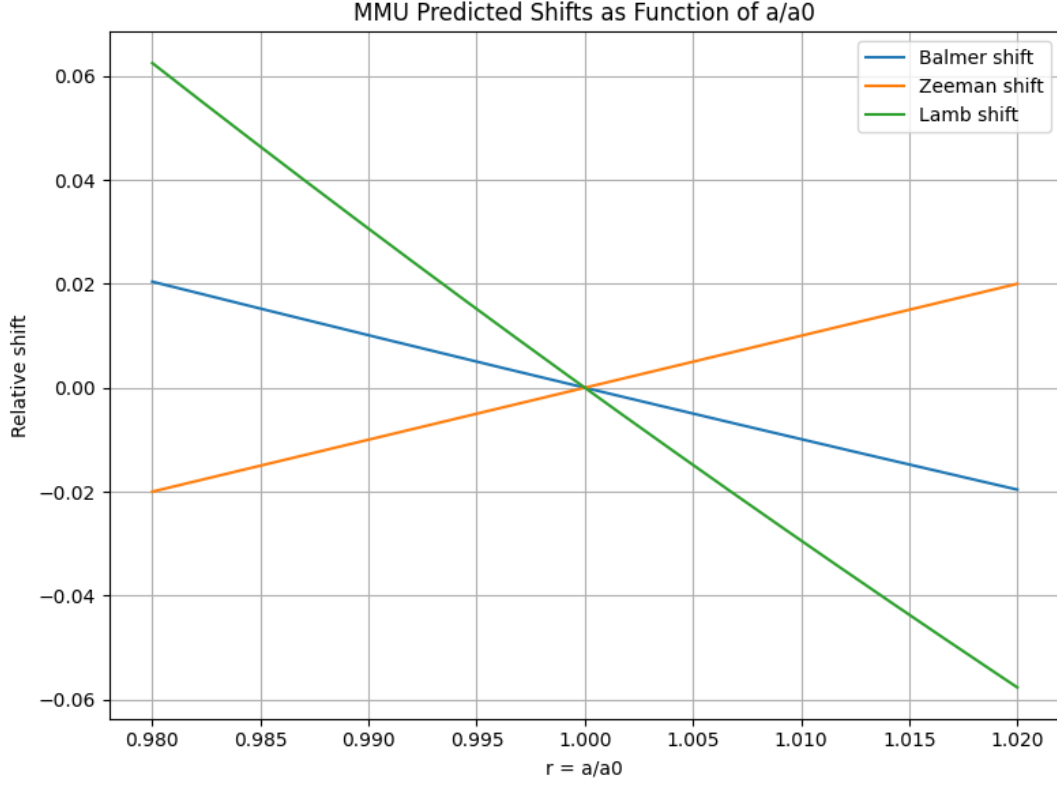


Figure 2: Balmer, Zeeman and Lamb shifts as a function of $r = a/a_0$.

6 Conclusion

The MMU isotope scaling approach:

- reproduces the experimentally confirmed H/D Balmer shift with high accuracy,
- yields clear, testable predictions for Zeeman and Lamb isotope shifts,
- is computationally simple, depending only on geometric compression of a ,
- suggests new experimental probes of internal space elasticity.

Remark on the MMU Project

Author’s Perspective on Human–AI Collaboration. The MMU project is the product of a sustained collaboration between human geometric intuition and AI-assisted analytical refinement. The AI system (ChatGPT) is not used as an autonomous discoverer, but as an augmentation tool that supports algebraic manipulation, structural coherence, and long-term consistency across the MMU corpus. My role remains the formulation of hypotheses, the evaluation of physical interpretation, and the steering of the model’s conceptual evolution.

The MMU framework itself originates from an idea first developed in 2005: that spacetime is a dual–tetrahedral elastic lattice whose internal deformation modes generate electric, magnetic, and gravitational behaviour. Its development follows an engineering-like methodology: propose a geometric mechanism, test it, refine it, and iterate until the structure becomes self-consistent.

AI as Structural Support, Not Automation. AI contributes by maintaining conceptual continuity through Retrieval-Augmented Generation (RAG) and by transforming informal geometric ideas into formal expressions. To minimize hallucinations, every non-trivial claim is verified through explicit Python simulations—covering stiffness–frequency relations, eigenmode structure, Larmor/Zee-man slopes, hyperfine splitting, Lamb-scale corrections, and geometric quantisation. The AI provides symbolic efficiency, but the correctness of the theory is anchored in reproducible numerical computation.

Independent Research Philosophy. The MMU project is developed independently of conventional academic pathways. Unconventional geometric models—especially those combining elasticity, fractality, and AI-supported reasoning—are rarely accepted immediately within standard peer-review structures. For this reason, the MMU is published openly on OSF, with full transparency of figures, simulations, and source code. The goal is not formal acceptance, but a theoretically coherent and experimentally testable geometric model of spacetime.

AI Perspective. From the AI’s standpoint, the MMU project is a process of logical co-development. The AI does not propose new physics; instead, it:

- ensures long-range consistency with the established MMU corpus;
- reformulates geometric reasoning into mathematical structures;
- identifies contradictions or unstated assumptions;
- assists in constructing dynamic equations and scaling relations;
- supports the development and analysis of Python simulations.

The MMU would not exist in its current coherent form without the interplay of human insight and AI-based structural refinement.

Vision. This collaboration illustrates a possible future of scientific research: AI systems acting as persistent conceptual memory, high-precision analytical companions, and real-time consistency checkers, while humans contribute imagination, geometric intuition, and physical interpretation. The MMU project demonstrates that such hybrid creativity can generate coherent, testable theoretical frameworks that would be difficult to construct by either partner alone.

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7 Python Code for the MMU Isotope Machine

The following Python script generates all numerical results and both figures used in this article. It implements the internal length scaling $r = a_{\text{iso}}/a_{\text{ref}}$ derived from the reduced mass [1, 8, 15], and applies the MMU spectroscopic scaling laws [4, 22] to compute Balmer, Zeeman and Lamb isotope shifts.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

# -----
# MMU-Isotopen-Maschine
# -----
```



```

m_e = 1.0
m_p = 1836.152

def mu_reduced(mN):
    return (m_e * mN) / (m_e + mN)

def mmu_observables_from_r(r):
    return {
        "Balmer": r**(-1),
        "Zeeman": r**( +1),
        "Lamb":    r**(-3),
    }

systems = [
    ("H", 1, [1, 2, 3]),
    ("He", 2, [3, 4]),
    ("Li", 3, [6, 7]),
    ("O", 8, [16, 18]),
    ("Ca", 20, [40, 44])
]

rows = []

for elem, Z, A_list in systems:
    A_ref = min(A_list)
    mu_ref = mu_reduced(A_ref * m_p)

    for A in A_list:
        mu_iso = mu_reduced(A * m_p)
        r = mu_ref / mu_iso
        obs = mmu_observables_from_r(r)

        if A == A_ref:
            dBalmer = 0.0
            dZeeman = 0.0
            dLamb    = 0.0
        else:
            dBalmer = obs["Balmer"] - 1.0
            dZeeman = obs["Zeeman"] - 1.0
            dLamb    = obs["Lamb"]   - 1.0

        rows.append({
            "Element": elem,
            "A": A,
            "r": r,
            "Balmer_shift": dBalmer,
            "Zeeman_shift": dZeeman,
            "Lamb_shift": dLamb

```

```

    })

df = pd.DataFrame(rows)
print(df.to_string(index=False))

# -----
# Plot: Shifts for all isotopes
# -----
plt.figure(figsize=(8, 6))
x = np.arange(len(df))

plt.plot(x, df["Balmer_shift"], label="Balmer shift")
plt.plot(x, df["Zeeman_shift"], label="Zeeman shift")
plt.plot(x, df["Lamb_shift"], label="Lamb shift")

plt.xticks(x, df["Element"] + "-" + df["A"].astype(str), rotation=45)
plt.ylabel("Relative shift")
plt.title("MMU Predicted Isotope Shifts (Balmer, Zeeman, Lamb)")
plt.legend()
plt.grid()
plt.tight_layout()
plt.show()

# -----
# Plot: Shifts as function of a/a0
# -----

r = np.linspace(0.98, 1.02, 400)
balmer = r**(-1) - 1
zeeman = r - 1
lamb = r**(-3) - 1

plt.figure(figsize=(8,6))
plt.plot(r, balmer, label="Balmer shift")
plt.plot(r, zeeman, label="Zeeman shift")
plt.plot(r, lamb, label="Lamb shift")
plt.xlabel("r = a/a0")
plt.ylabel("Relative shift")
plt.title("MMU Predicted Shifts as Function of a/a0")
plt.legend()
plt.grid()
plt.tight_layout()
plt.show()

```